Fiscal Year:	FY 2010	**Task Last Updated:**	FY 09/13/2010
PI Name:	Mitchell, Cary Ph.D.		
Project Title:	(NSCORT) for Advanced Life Support at Purdue/Howard/AAMU - Solid Waste Resource Recovery		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	West Lafayette	**State:**	IN
Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSCORT
Start Date:	12/01/2002	**End Date:**	11/30/2009
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2009 per PI (1/09) NOTE: Received NCE to 11/30/2008 per PI (11/07)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Alleman , Jim (Purdue Univesity) Glass, Charles (Howard University) Turco, Ron (Purdue University) Nies, Larry (Purdue University) Mitchell, Cary (Purdue University)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	PAABLO (plant-based aerobic-anaerobic bioreactor linked operation) A significant fraction of waste material (plant residue, greywater and human waste) generated during space missions and on the earth, contains recoverable energy. The specific objective of the project is to optimize an approach that will demonstrate a Plant-based Anaerobic-aerobic Bioreactor Linked Operation (PABLO) that efficiently converts plant residuals and greywater into fuel grade methane (CH4), electricity, and biosolids that will be suitable for plant and fungal culture. This effort will optimize the potential for CH4 production by aerobically treating anaerobically recalcitrant waste materials - biosolids before they are used as a plant growth material. This step will breakdown the resistant materials, freeing soluble carbon and stabilizing the materials. In a second part of the project we will modified the PABLO system to a system concerned with converting human waste in to CH4 and biosolids. The Waste-ABLO (WABLO) approach is for the conversion of human waste into CH4 and biosolid residuals that are condensed but not used in the production of food. In both the space based and Earth-based systems the starting point will be waste materials. The major difference is in the space-based system we will be dealing primarily with dried residues that will be rewetted with greywater. In the earth-based system, the residues will be wet or dry. Outputs from the system will include: CH4, CO2 and solids. In this system, the residuals will be treated with an aerobic process to and returned to the anaerobic system. The residuals that settle in the system will be removed and land applied (earth) or used in confined growth systems (Mars). The project is divided into three tasks: Task 1. The objective is use an anaerobic-aerobic bioreactor system to optimize production of methane from plant-based waste materials and for the recovery of high-quality biosolids. Task 2. Methods for the optimization of the PABLO biosolids for use in fungal and plant cultivation will be investigated. This will require an evaluation of the makeup the aerobic reactor's microbial population to maximize the formation of soluble carbon for use in the anaerobic portion of PABLO. Task 3. The objective is use a bioreactor system to optimize production of methane from human waste materials
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Further development of biological methods for converting organic wastes into fuels, chemicals, and other useful products. Gain a greater understanding for recycling ammonia-rich wastewater streams. The PABLO system is designed to recycle waste materials from other important processes here on Earth. Extra crop residues are mixed with greywater and a fungi to produce mushrooms. This allows for the breakdown of the residues, recycling of greywater and production of food. Degraded residues and more graywater are further utilized by the addition of anaerobic digester sludge. This combination produces carbon dioxide and methane. The methane can be used as a fuel source alone or further reformed to produce hydrogen for use in fuel cells for power generation.
Task Progress:	For the first set of biological methane production (BMP) experiments, corn residue was milled and added to a phosphate buffer. Anaerobic digester sludge was used as an inoculum. Digesters were incubated at 23oC and periodically checked for gas and acid production over a period of time. Corn residue in a phosphate buffer showed poor methane production over a 45 day time period. The maximum amount of methane produced was 0.09mLs after 32 days. We also found this trend of low methane production to be true for several other crop residues that were tested. Low methane production resulted from the lignin matrix surrounding cellulose carbon sources, preventing the anaerobic bacteria from accesses these sources (Muller and Trosch, 1986). Because of these findings, an efficient and simple pretreatment method should be used on crop residues to reduce the lignin content. There are several methods of pretreatment that could be employed to break down lignin including gamma-ray irradiation, alkali treatment, or aerobic composting (Delgenes, et al, 2002). All of these methods would either require hauling large amounts of chemicals or supplies, or consume valuable resources such as oxygen. Lignin degrading fungi appear to be a promising method of pretreatment for space missions. The mushroom species Pleurotus ostreatus, or Oyster mushroom, is edible, easy to grow, and is able to selectively degrade lignin, leaving the more soluble forms of carbon available for digestion by anaerobic microorganisms (Muller and Trosch, 1986). This species also grows well in a high carbon dioxide environment during its longest growth stage, so oxygen consumption could be minimized with careful planning (Stamets, 2000). The oyster mushroom requires a lot of water to sustain good growth; however, we have found that this mushroom species is able to grow well when using a mimicked hygiene graywater solution in place of regular water (Hanford, 2004). The process of growing the Oyster mushroom is quite simple. The mycelium is first grown on a grain, such as wheat or rye, to establish the fungi. Plant materials to be degraded are then pasteurized. Once cooled, the grain grown mycelium are then transferred and mixed with the pasteurized plant materials and grown under high humidity conditions. While the mycelium is being established, they grow best under a higher carbon dioxide concentration. Oyster mushrooms will begin to appear after about 18 days and grow to fruition in just 3-5 days (Stamets, 2000). Graywater can be used throughout the mushroom growing process to conserve drinking water. After one harvest of mushrooms grown on pasteurized corn residue, graywater was leached through the remaining residue and collected for methane production studies. This time, only the leached graywater and anaerobic sludge was used. No buffer was added. The digesters were incubated at 35oC. Methane production was greatly increased over digested corn residue in buffer, with a maximum production value of 8.3mLs after 35 days. Milled plant residue alone will not produce enough methane to use a fuel source. A pretreatment method should be used to break down the lignin. The Oyster mushroom is an efficient and simple pretreatment method for degrading the lignin found in plants. This method opens up useable forms of carbon to the anaerobic bacteria and greatly increases the amount of methane produced. Using graywater throughout the methane production process will conserve valuable drinking quality water while recycling another waste material created by the crew on space missions. Degrading plant materials anaerobically is showing great promise for both long term space missions and here on Earth.

	During 2009, the above research was expanded to two other residues; miscanthus and tomato plant material. The graphs below show a comparison of the methane produced from three residues treated with fungi at each stage in the method. Of the three residues tested, miscanthus produced the least amount of methane during all stages. More methane was produced after pasteurization and at the pinhead stage, but there was little difference between post-pasteurization and the pinhead stage. Corn residue produced more methane after pasteurization and after the pinhead stage, reaching a maximum methane production of 8.6mL and 8.3mL respectively. Tomato residue reached maximum methane production of all treatments at 9.5mL at the pinhead stage. Uninoculated controls were used for all treatments and consistently showed less methane production than inoculated treatments. This suggests that fungi are an effective method of pretreatment for production of methane from crop residues. Further research is needed to optimize this system. Greater methane production may be possible by extending the amount of time the bottles are incubated for. The bottles used for the experiments were small and only held a limited amount of gas. A different gas collection system should be developed so that pressure does not build up in the bottles.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Abstracts for Journals and Proceedings	Mitchell C, Whitaker D, Banks MK, Heber AJ, Turco RF, Nies LF, Alleman JE, Sharvelle SE, Li C, Heller M. "Microbial Bioreactor Development in the ALS NSCORT. " 37th COSPAR Scientific Assembly, Montreal, July 13-20, 2008. 37th COSPAR Scientific Assembly, Montreal, 13-20 July 2008. Program and Abstracts, p.2062. , Jul-2008
Articles in Peer-reviewed Journals	Turco R, Heller M. "Development of a plant based anaerobic-aerobic bioreactor linked operation for fuel production on Mars." (In Progress), September 2010. , Sep-2010